

TI-DeepONet: Learnable Time Integration for Stable Long-Term Extrapolation

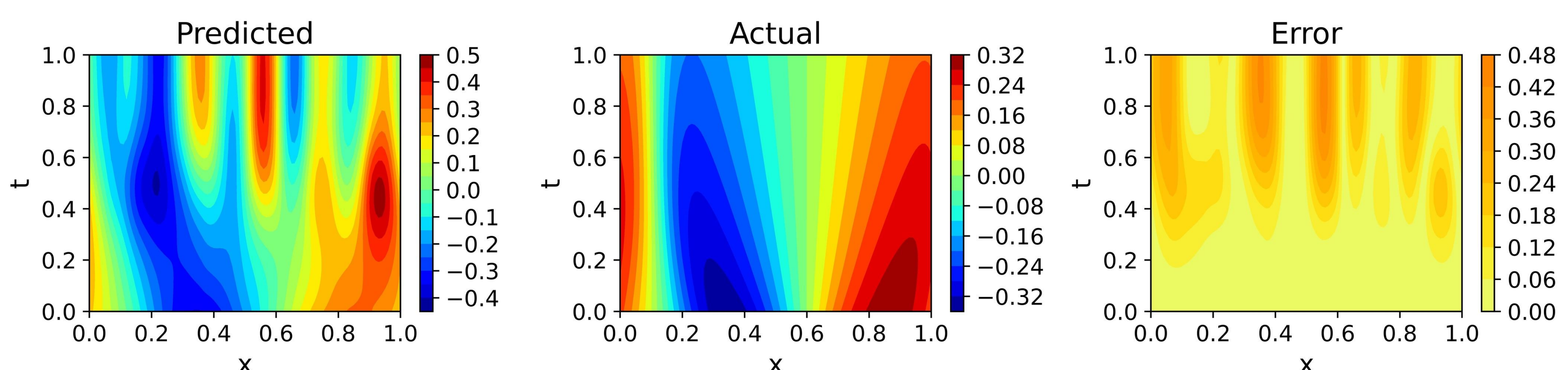
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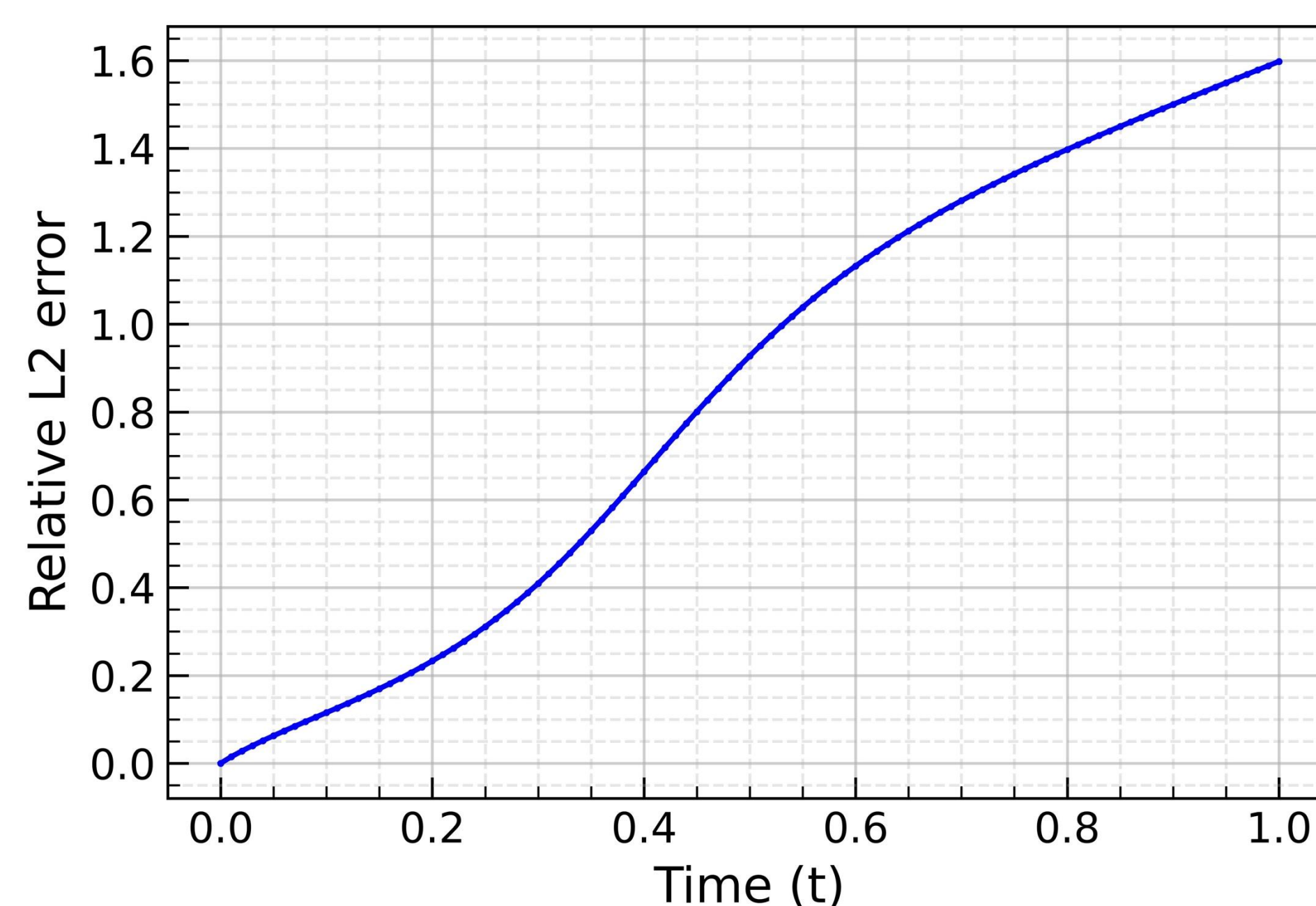
Motivation

- Traditional numerical methods, while accurate, are prohibitively expensive for large-scale simulations.
- Neural Operator (NO)-based surrogate models have gained popularity for mapping between infinite-dimensional input and output function spaces without the need to re-run numerical simulations for new parametric conditions.
- NOs operate under two modes – **full rollout (FR)** and **autoregressive (AR)**.
- FR**: Fixed basis functions and coefficients (non-Markovian structure).
- AR**: Recursive predictions in time (error accumulation).

Our vision: Develop a computationally-efficient and physically-consistent neural operator surrogate model that can respect the underlying causal structure in time-dependent PDE systems



- In an autoregressive setting, the Markovian structure is preserved, but there is significant error accumulation.
- For example, in a canonical 1D Burgers' PDE, the error can reach up to 160% at the final timestep.
- Error propagation occurs due to prediction on potentially noisy inputs from previous timesteps.



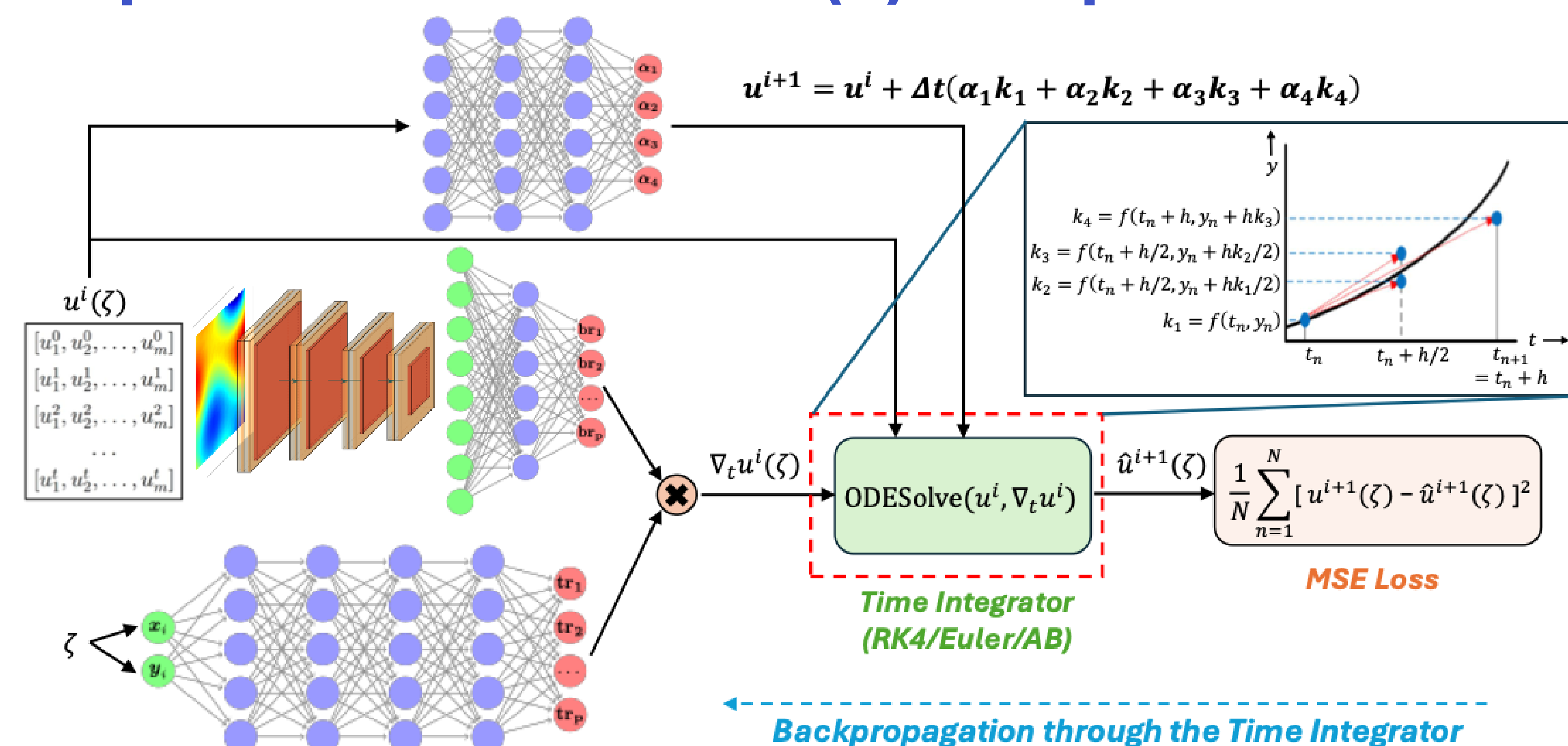
Challenges

- The main goal is to mitigate the error accumulation of vanilla neural operators for time-dependent systems.
- Additionally, in realistic scenarios, ground truth is available only at limited timesteps within the simulation domain. We aim to enable extrapolation to temporal regions not seen during training and to predict or forecast long-term dynamics.

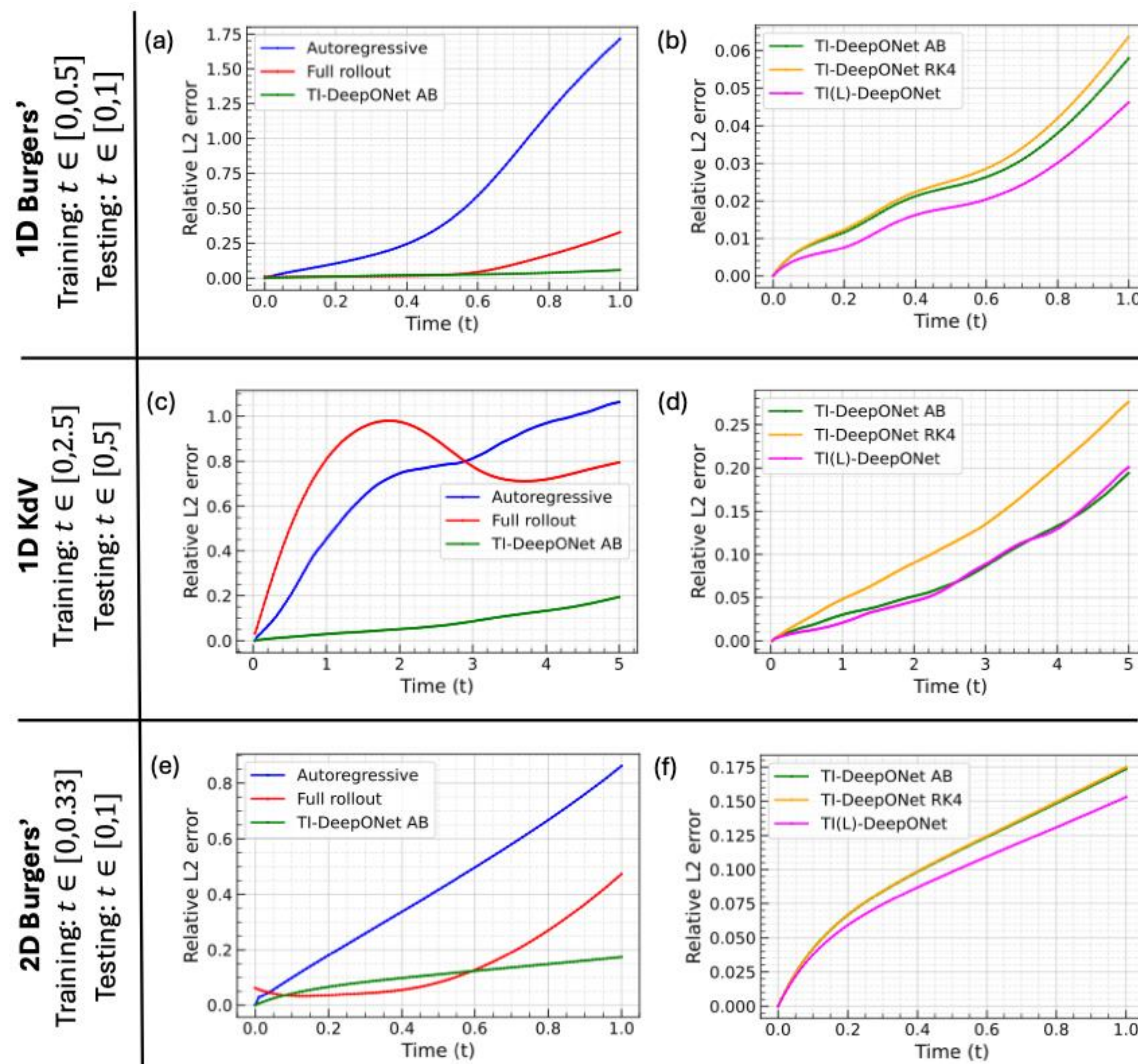
Proposed solution

- The proposed **TI-DeepONet** approach reformulates the learning task by modeling the instantaneous time-derivative operator.
- We also propose an enhanced version, **TI(L)-DeepONet**, which performs numerical time integration using an adaptive RK4 update.
- This approach achieves a significant reduction in L_2 extrapolation errors: 81% compared to autoregressive rollouts and 70% compared to full rollouts.
- Examples considered: (1) 1D Burgers', (2) 1D KdV, and (3) 2D Burgers'.

Proposed framework: TI(L)-DeepONet



Results



References

- Nayak and Goswami, "TI-DeepONet: Learnable Time Integration for Stable Long-Term Extrapolation." *arXiv preprint arXiv:2505.17341* (2025).
- Lu, Lu, et al. "Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators." *Nature machine intelligence* 3.3 (2021): 218-229.